

Question Number	Answer	Additional guidance	Mark
7(a)(i)	The only correct answer is A Q <i>A is not correct because Q transports blood to renal vein</i> <i>B is not correct because Q transports blood to renal vein</i> <i>D is not correct because Q transports blood to renal vein</i>		(1)

Question Number	Answer	Additional guidance	Mark
7(a)(ii)	The only correct answer is A P, Q and U <i>B is not correct because only P and Q would be found in the medulla region</i> <i>C is not correct because only P and U would be found in the medulla region</i> <i>D is not correct because only P and U would be found in the medulla region</i>		(1)

Question Number	Answer	Additional guidance	Mark
7(a)(iii)	An answer that includes the following points: • orientation described (1) • {structure A / brush border / microvilli} gives increased surface area (1) • (more) {structure B / mitochondria} to supply ATP (1)	ACCEPT {brush border / structure A} is in the lumen of the tubule ACCEPT microvilli all along {lining of the PCT / apical side} ACCEPT {mitochondria / structure B} are located {throughout / at the base cell / basal side} ignore incorrect name / structure ACCEPT {structure B / mitochondria} for {aerobic respiration / releasing energy / producing ATP} ignore producing energy	(3)

Question Number	Answer	Additional guidance	Mark
7(b)	An answer that includes four of the following points: • mean volume of urine production peaks at {60 minutes / one hour} (1) • mean concentration of ions is lowest at {60 minutes / one hour} (1) • mean mass of urea peaks at {60 minutes / one hour} (1) • at 150 minutes {mass of urea / volume of urine / concentration of ions} was same as {at 0 mins / the start / returns to normal} (1) • mean mass of urea is greatest when volume of urine greatest (1) • there is no indication of {sample size / gender / health issues / hydrated prior to test} (1)	ACCEPT increases up to 60 mins ACCEPT concentration of ions decreases up to 60 mins ACCEPT mass of urea increases up to 60 mins ignore mass of urine ACCEPT converse ACCEPT other correct linkages e.g. highest mean volume of urine is when mean concentration of ions in urine is least / as the volume of urine increases the concentration of ions decreases / / mass of urea is lowest when concentration of ions greatest ACCEPT there are no {error bars / standard deviation} so unable to determine significance of the results	(4)

Question Number	Answer	Additional guidance	Mark
7(c)	An explanation that includes three of the following points: • ADH binds to receptors {on distal convoluted tubules/ collecting ducts} (1) • (causing) aquaporin channels to move (from cytoplasm) into the (apical) membrane of the tubules (1) • (allowing) water to be reabsorbed out of {the collecting ducts / distal convoluted tubules} (1) • (the water) is reabsorbed into blood / the blood (sodium) ion concentration reduces (1)	ACCEPT water channels for aquaporins ACCEPT more {aquaporin / water channels} in membrane ACCEPT {collecting duct /distal convoluted tubules} becomes more permeable to water ACCEPT the blood {becomes more dilute / has increased water potential}	(3)

